Curriculum Vitae

Mattia Puddu

Date and place of birth:	January 29, 1996 – Nuoro, Italy
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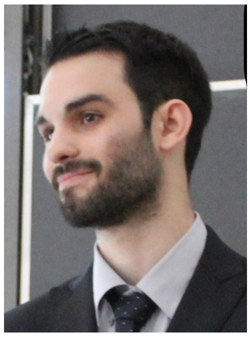

Math interests

My current research focuses on extending the theory of Thomas decomposition from algebraic and differential systems of polynomial equations and inequations to the difference-algebraic setting.
More broadly, within difference and differential algebra, I am also interested in the Galois theory of linear difference and differential equations.
Beyond these topics, my mathematical focus includes modular forms, classical and difference algebraic geometry, and representation theory, with particular emphasis of the latter to these areas.

Employment

2025 – **PhD Student in Mathematics**, RWTH Aachen University, Germany
Project: Effective Difference Algebra and Difference Algebraic Groups
Advisors: Prof. Daniel Robertz and Priv.-Doz. Dr. Annette Bachmayr

Education

2019 – 2025 University of Pisa, **Master's Degree in Mathematics**.
Final grade: 110/110 *summa cum laude*
Thesis: *The Differential Galois Group of the Family of All Regular Singular Differential Equations over* $\mathbb{C}(z)$.

Advisor: Prof. Tamás Szamuely
Examiner: Prof. Andrea Maffei
Available at: https://mattiapuddu25.github.io/MasterThesis.html

2015 – 2019 University of Pisa, **Bachelor's Degree in Mathematics**.
Final grade: 110/110 *summa cum laude.*
Thesis: *The Picard-Vessiot Theory – An Introduction with Applications to Fuchsian Differential Equations.*

Advisor: Prof. Paolo Acquistapace.
Available at: https://mattiapuddu25.github.io/BachelorThesis.html

2010 – 2015 Scientific High School "Enrico Fermi", Nuoro, Italy.
Final grade: 100/100 *summa cum laude.*

Conferences, Workshops and Summer Schools

May 2026 DART 2026 — Differential Algebra and Related Topics XIV.
CNRS Center La Vieille Perrotine, Île d'Oléron, France, May 24–30, 2026.

Poster: Normal form algorithm for polynomial membership and quasi-simple difference systems.
https://magix.lix.polytechnique.fr/magix/odelix/dart26/dart26.html

Sep 2025 Summer School "Computer Algebra with OSCAR".
Pfalzakademie, Lambrecht, Germany, September 15–19, 2025.

https://www.oscar-system.org/events/meetings/2025-09/

Teaching activity

2023 University of Pisa — Department of Civil and Industrial Engineering
Tutor for the pre-course *Matematica 0* (Prof. Irene Venturi)

- Held weekly online sessions to help students pass the OFA admission tests

2022 University of Pisa — Department of Agricultural, Food and Agro-Environmental Sciences
Teaching assistant for *Matematica e Statistica* (Bachelor's in Viticolture and Oenology, Prof. Valentino Magnani)

- Conducted the "Mathematics 0" pre-course, including preparation of exams and corresponding solutions
- Held weekly in-class meetings with students for problem-solving and theoretical clarifications
- Reviewed exam papers and solutions from midterm tests throughout the semester
- Supervised students during the exams mentioned above

2021 – 2022 University of Pisa — Department of Computer Science
Tutor for *Algebra Lineare B* (Bachelor's in Computer Science, Prof. Mauro Di Nasso)

- Held meetings with students (mostly upon request) for problem-solving and theoretical clarifications

2019 – 2020 University of Pisa — Department of Physics
Teaching assistant for *Geometria A* (Bachelor's in Physics, Profs. Mario Salvetti and Filippo Di Santo)

- Held meetings with students (mostly upon request) for problem-solving and theoretical clarifications
- Corrected midterm and final exams, including the preparation of written solutions

Computer Skills

Operating systems: Windows, macOS, Linux, Android
Software/languages: Matlab, C, Julia, HTML (basic), LaTeX (expert)

Language Skills

Italian: native
English: fluent
Spanish: independent user

Selected Elective Courses

Completed during the Master's and Bachelor's degrees at the University of Pisa, in addition to the mandatory curriculum.

1. **Coxeter Groups** (geometric realization and combinatorial characterization; Bruhat order; root systems; classification of finite Coxeter groups; applications in combinatorics and topology) with a grade of 30/30 *cum laude*
(taught by Profs. Michele D'Adderio and Mario Salvetti)
2. **Elements of Algebraic Geometry** (projective geometry; Bezout's theorem; Nullstellensatz; affine and quasi-projective varieties; Zariski topology; irreducible decomposition; morphisms and rational maps; dimension; tangent spaces and singular points; Segre, Veronese, and Grassmannian varieties) with a grade of 30/30 *cum laude*
(taught by Prof. Rita Pardini)
3. **Algebraic Geometry C** (Riemann surfaces; holomorphic and meromorphic functions; group actions and Hurwitz bound; differential forms; divisors and linear systems; Riemann-Roch theorem; Serre duality; Weierstrass points; Jacobian variety; theorem of Abel and Jacobi) with a grade of 30/30 *cum laude*
(taught by Prof. Marco Franciosi)
4. **Algebraic Combinatorics** (enumerative and algebraic combinatorics; ordinary and exponential generating functions; Lagrange inversion; symbolic methods; transfer matrix method; MacMahon master theorem; symmetric functions) with a grade of 30/30 *cum laude*
(taught by Prof. Michele D'Adderio)
5. **Analytic Number Theory A** (entire functions of finite order; Gamma function and Stirling's formula; Riemann Zeta function; Riemann-von Mangoldt formula; prime number theorem and de la Vallèe Poussin theorem; Mellin transform; characters and Dirichlet L-functions; primes in arithmetic progressions) with a grade of 30/30 *cum laude*
(taught by Prof. Giuseppe Puglisi)
6. **Groups and Representations** (representations of finite groups; complete reducibility; Schur's lemma; characters and orthonormality; induced representations; representations of semisimple algebras; representations of the symmetric group; Schur's duality; representations of the general linear group) with a grade of 30/30 *cum laude*
(taught by Profs. Michele D'Adderio and Giovanni Gaiffi)
7. **Elements of Algebraic Topology** (singular homology; CW-complexes; cellular homology; cohomology ring and cup product; Poincaré duality; homotopy groups; cellular approximation; homotopy groups of spheres) with a grade of 30/30 *cum laude*
(taught by Prof. Filippo Callegaro)
8. **Elements of Complex Analysis** (Hurwitz theorems; Schwarz lemma; automorphisms of the disk and the Riemann sphere; Poincaré distance; Wolff-Denjoy theorem; Riemann uniformization theorem; several complex variables; Hartogs extension theorem; plurisubharmonic functions; Levi problem) with a grade of 30/30 *cum laude*
(taught by Prof. Marco Abate)
9. **Modular Forms** (the modular group $\mathrm{SL}_2(\mathbb{Z})$; classical modular forms; Fourier expansions; Eisenstein series; theta series and representations as sums of squares; congruence subgroups; Hecke operators; Petersson inner product; L-functions attached to eigenforms; Euler product decomposition) with a grade of 30/30 *cum laude*
(taught by Profs. Andrea Maffei and Davide Lombardo)
10. **Lie Algebras and Lie Groups** (semisimple, nilpotent, and solvable Lie algebras; Killing form; Weyl's theorem; representations of $\mathfrak{sl}(2, \mathbb{K})$; weight lattice; root systems; Dynkin diagrams and classification; Poincaré-Birkhoff-Witt theorem; Serre relations; Lie groups; Baker-Campbell-Hausdorff formula) with a grade of 30/30 *cum laude*

(taught by Profs. Giovanni Gaiffi and Enrico Le Donne)

11. **Algebraic Number Theory 1** (integral extensions; trace and norm; ring of integers; Dedekind domains; fractional ideals and class group; Kummer's theorem; ramification and inertia in Galois extensions; decomposition and inertia groups; Frobenius morphism; Minkowski convex body theorem; finiteness of the class group; Dirichlet unit theorem) with a grade of 28/30
(taught by Profs. Ilaria Del Corso and Roberto Dvornicich)
12. **Elementary Number Theory** (prime number theorem; Lagrange's four-squares theorem; quadratic residues and reciprocity law; multiplicative groups modulo q; arithmetic functions; Dirichlet series; Mertens and Chebychev theorems; Dirichlet's theorem on primes in arithmetic progressions) with a grade of 30/30 *cum laude*
(taught by Prof. Giuseppe Puglisi)

Other Activities

Mathematical Olympiads. I have always been interested in the Italian Mathematical Olympiad project. Over the years, I compiled my solutions to the National Team Competitions held in Cesenatico into two files, which also include theory notes useful for competition preparation.

Available at: https://mattiapuddu25.github.io/MathOlympiads.html